BTL1213 Operating Systems / BIT1213 Operating Systems / BCL1213 Operating Systems

SEGi University

Chapter 03 Memory Management - Part A



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

BACHELOR OF COMPUTER SCIENCE (HONS)

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Learning Objectives:

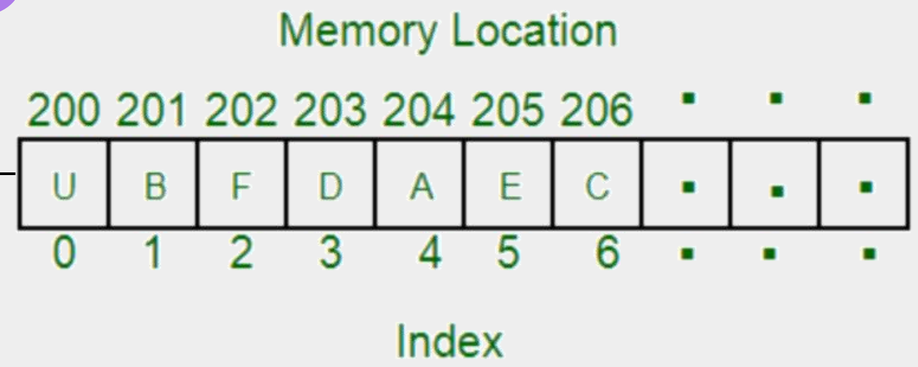
By the end of this lesson, students will be able to:

- Understanding concept of memory and it’s important role.
- Identify different types of memory.
- Analyze examples of memory allocation.
- Explain the factors affecting memory speed.

OVERVIEW

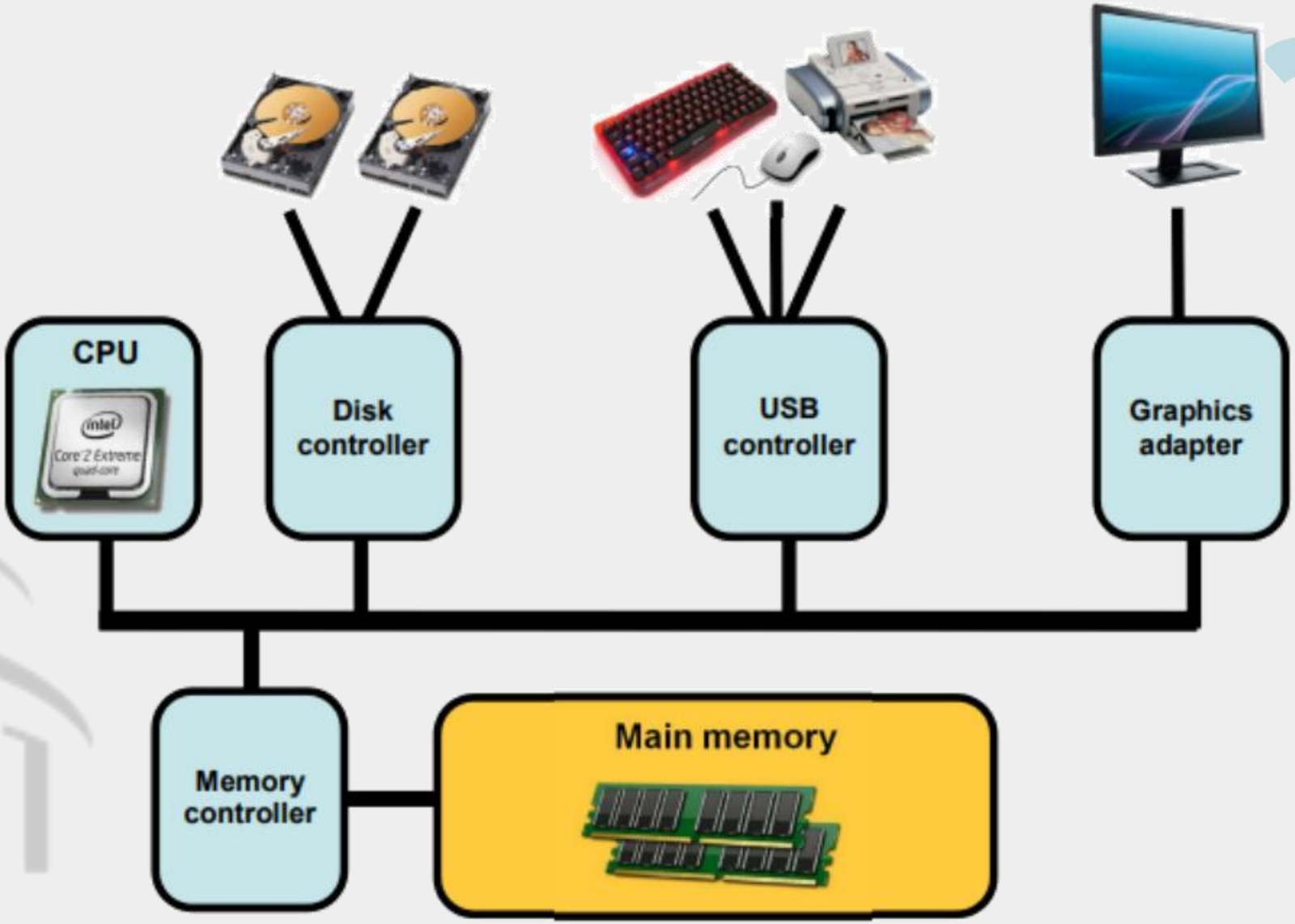
Topic for today's main discussion:

- Definition of Memory
- What is Memory Management ?
- Memory Hierarchy
- Types of Memory Allocation
- Memory Access in Operating System



Introduction

- Computers have different types of memory, which our focus is RAM (stands for Random Access Memory) because that is where the OS manages memory for running programs.
- The main memory is central of a Modern Computer.
- Main memory is associated with the processor, so moving instructions and information into and out of the processor is extremely fast.


CPU
Disk controller
USB controller
Graphics adapter
Memory controller
Main memory

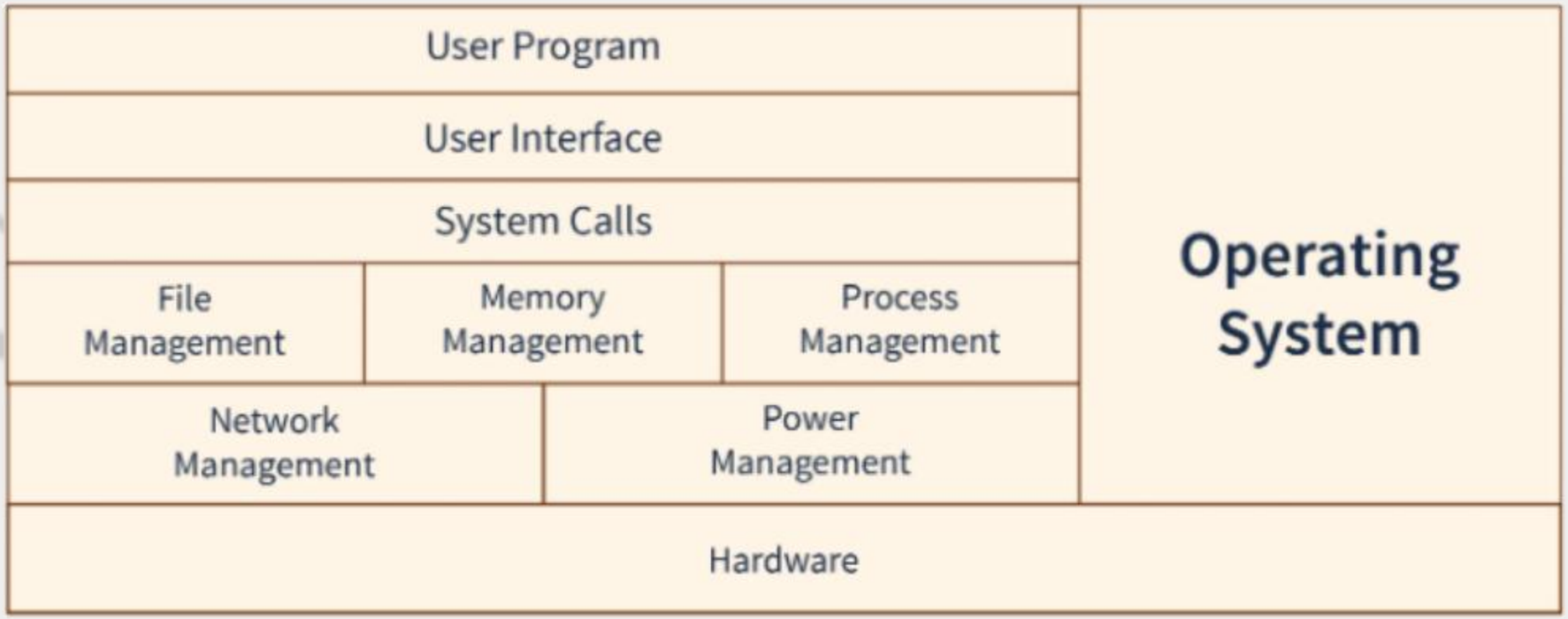
01 Definition of Memory

- Memory in a computer is like a temporary storage space where data and instructions are kept while the computer is working..
- Used to store instructions and process data.
- The memory comprises a large array or group of words or bytes, with its own location.
- The primary purpose of a computer system is to execute programs. So, programs, should be in the main memory during execution.

02 What is Memory Management?

- Memory management in operating system is the process of controlling and coordinating computer memory.
- It is used for achieving better concurrency, system performance, and memory utilization.

- Memory management moves processes from primary memory to secondary memory and vice versa. It also keeps track of available memory, memory allocation, and unallocated.



Why Memory Management?

Why use Memory Management in OS?

- To keeps track of the status of memory location.
- To achieve concurrency, system performance and memory utilization.
- To addresses the system’s primary memory.
- To protect the memory allocated from being corrupted.
- To enables sharing of memory spaces among processes.

How Memory Management works?

- Memory management in operating system is divided into certain steps to focus on each process carefully and effectively.
    a) Memory Allocation (Assigning Space)
    b) Memory Deallocation (Freeing Space)
    c) Memory Protection (Preventing Conflicts Between Programs)
    d) Memory Organization (Structuring Memory Efficiently)

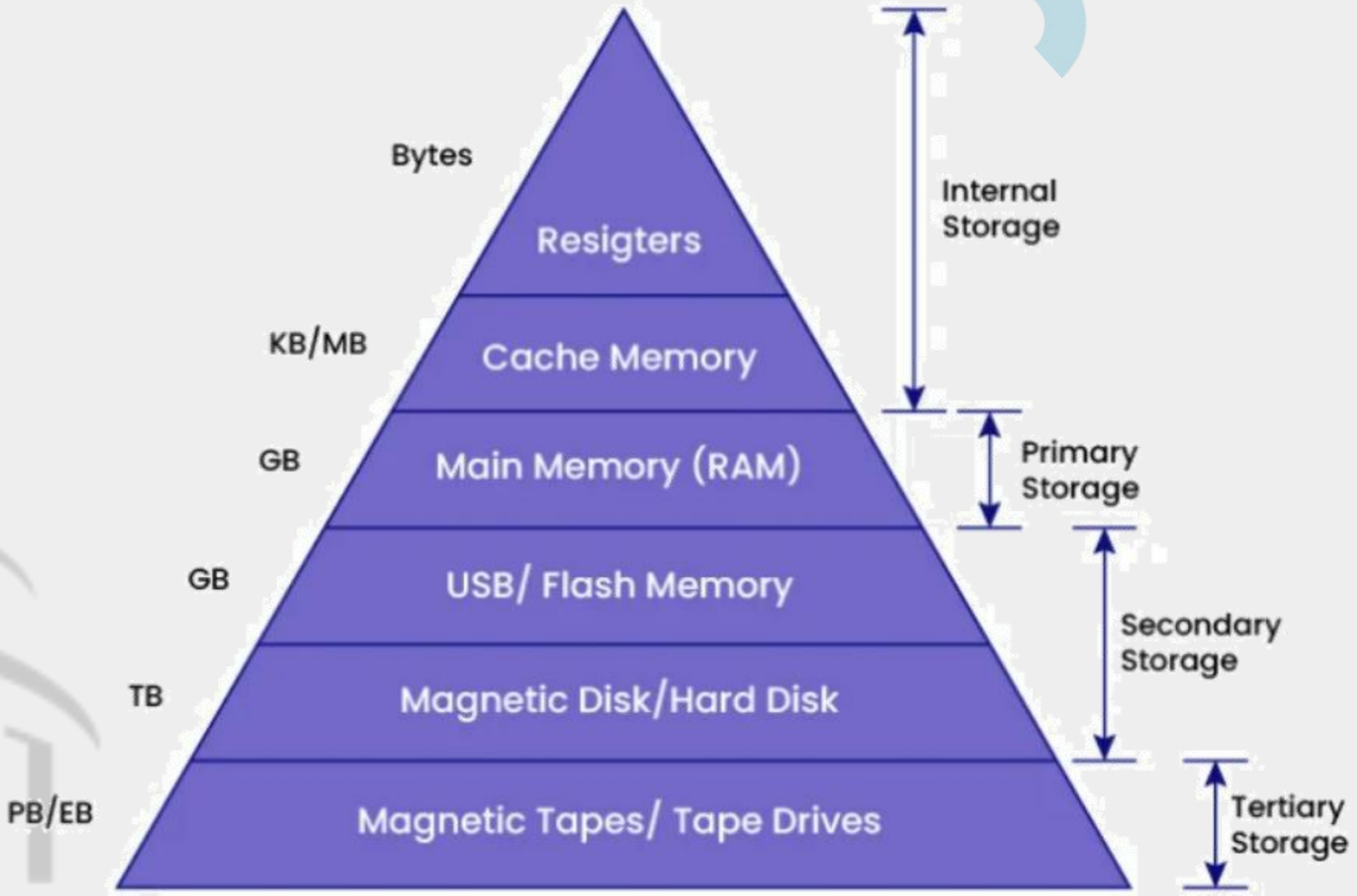
03 Memory Hierarchy

- The memory hierarchy in a computer system is a structured organization of storage layers, each with varying access speeds, capacities, and costs.
- It is based on a program behavior that is known as locality of references.
- It helps to manage memory by organizing the various types of data into meaningful containers.
- This design aims to balance speed, capacity, and cost effectively.

- 2 Types of Memory Hierarchy:

  a) External Memory or Secondary Memory: Comprising of Magnetic Disk, Optical Disk, and Magnetic Tape.

  b) Internal Memory or Primary Memory: Comprising of Main Memory, Cache Memory & CPU registers.


Bytes
Resigters
KB/MB
Cache Memory
GB
Main Memory (RAM)
GB
USB/ Flash Memory
TB
Magnetic Disk/Hard Disk
PB/EB
Magnetic Tapes/ Tape Drives
Internal Storage
Primary Storage
Secondary Storage
Tertiary Storage

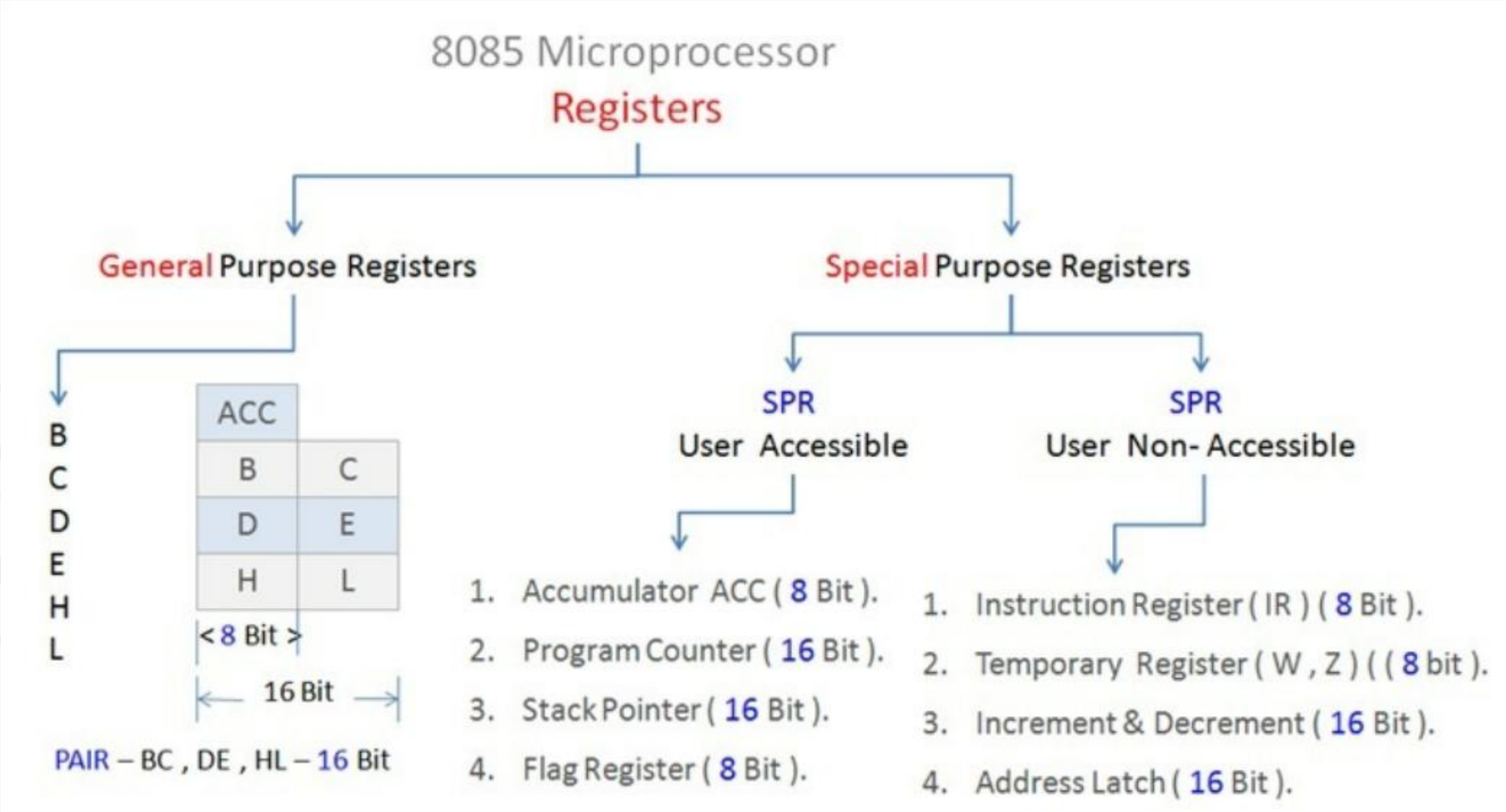
Can we just use cache memory or RAM only?



1. CPU Registers

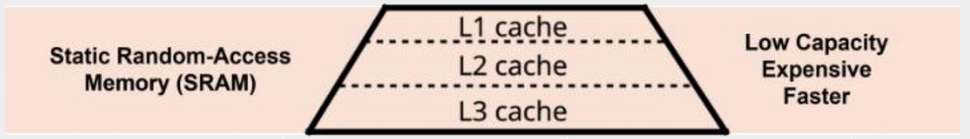
Feature	Details
Location	Built inside the CPU
Purpose	Stores instructions, data, and addresses currently being processed
Speed	Fastest memory in the computer
Capacity	Usually less than 1 KB
Access Time	Approximately 0.2–1 nanosecond
Volatility	Volatile
Cost per Bit	Highest
Common Registers	Program Counter (PC), Instruction Register (IR), Accumulator, Memory Address Register (MAR), Memory Data Register (MDR), General Purpose Registers (GPRs)
Example	During an addition operation, operands are first loaded into CPU registers before execution.

- Example of different types of registers used in Intel 8085 Architecture.



2. Cache Memory

Feature	Details
Location	Between the CPU and Main Memory (RAM)
Purpose	Reduces CPU access time by storing frequently used data
Speed	Very Fast
Capacity	Small (larger than registers but smaller than RAM)
Volatility	Volatile
Cost per Bit	Very High
Types	L1 Cache: Fastest and smallest, inside each core. L2 Cache: Larger than L1 and slightly slower. L3 Cache: Shared among CPU cores, the largest cache, and faster than RAM.
Important Terms	Cache Hit: Requested data is found in cache. Cache Miss: Requested data is fetched from RAM because it is not available in cache.
Example	Frequently used application data is stored in a cache for faster processing.



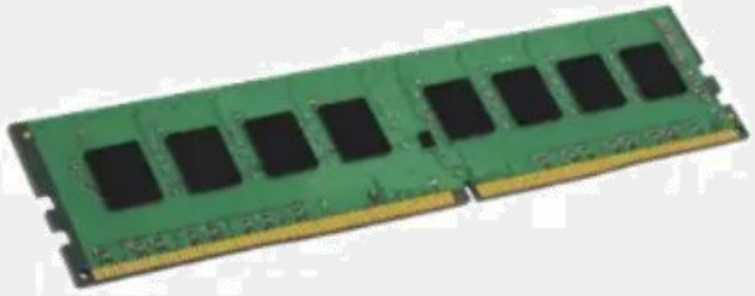
3. Main Memory (RAM)

Feature	Details
Location	Connected directly to the motherboard
Purpose	Stores active programs and data currently used by the CPU
Speed	Fast
Capacity	Medium
Volatility	Volatile
Cost per Bit	Moderate
Types	SRAM: Faster, expensive, mainly used in cache.DRAM: Slower, economical, used as main memory.
Modern Technologies	DDR4, DDR5
Example	Applications like Chrome or MS Word load into RAM while running.

- Types of Main Memory

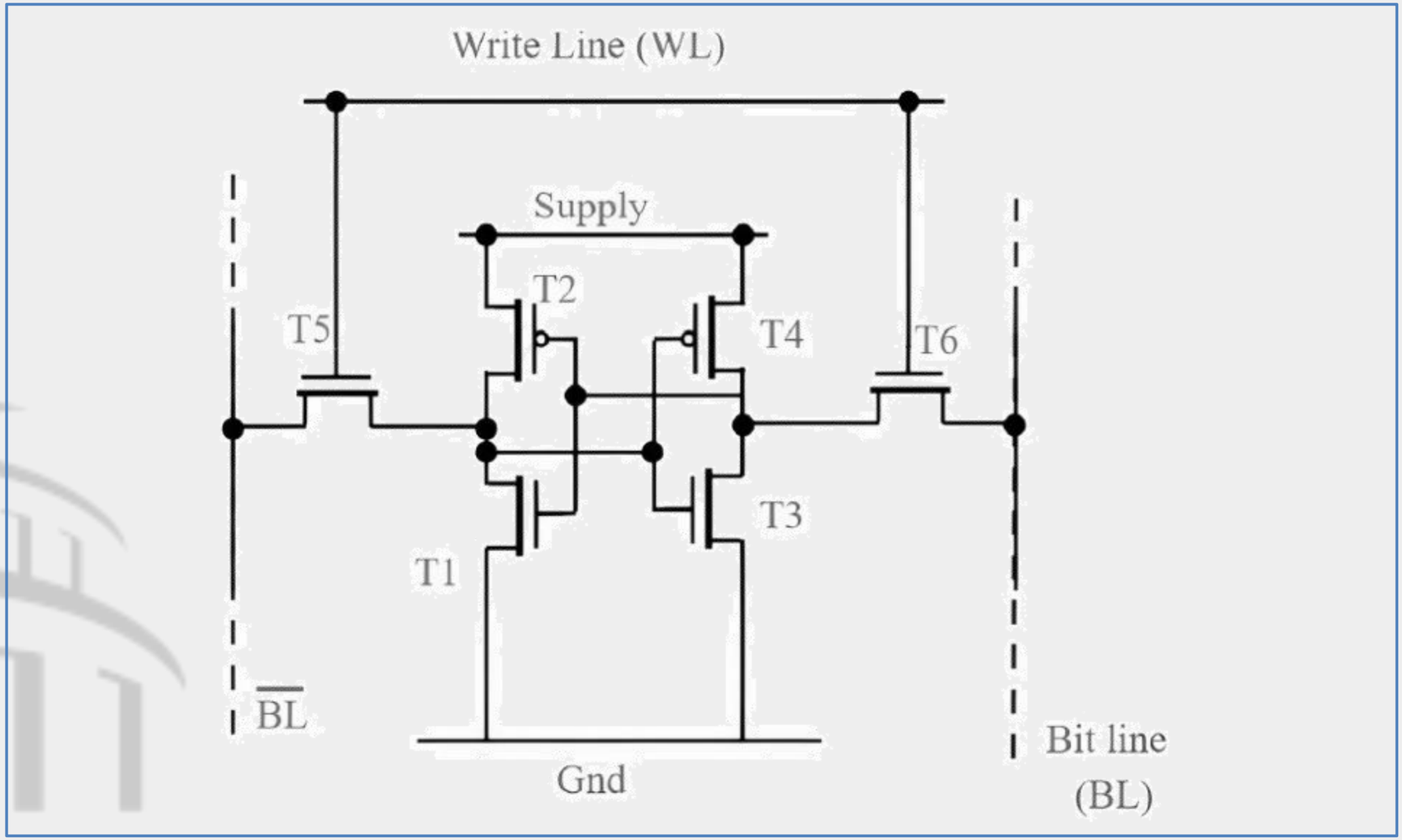


STATIC RAM (SRAM)

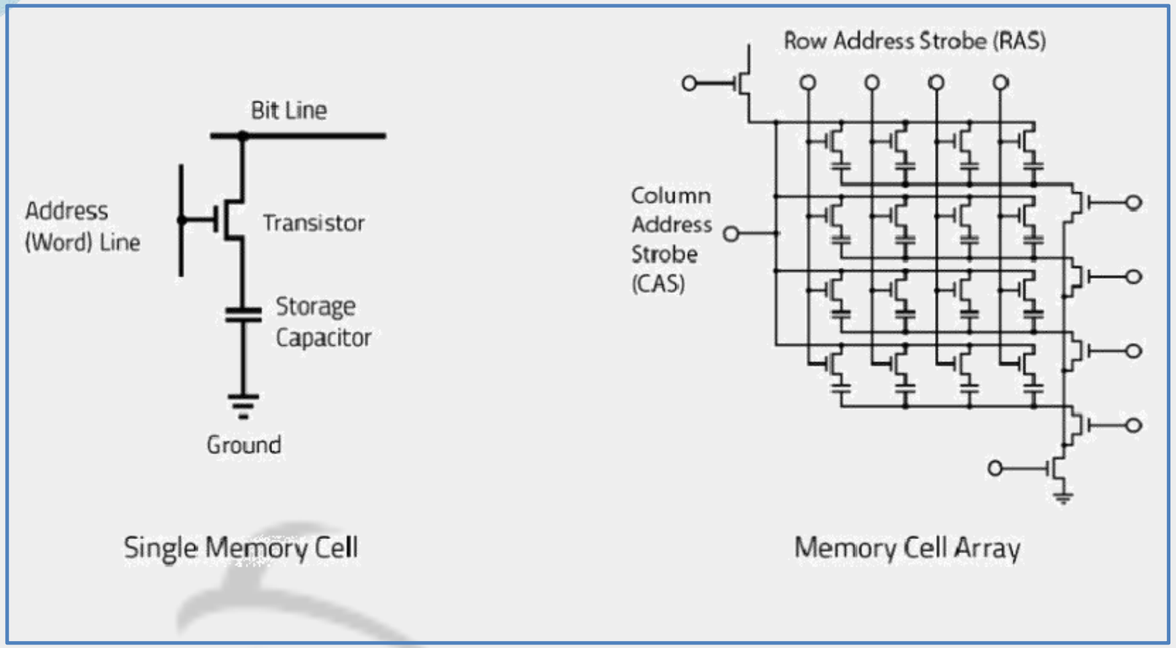


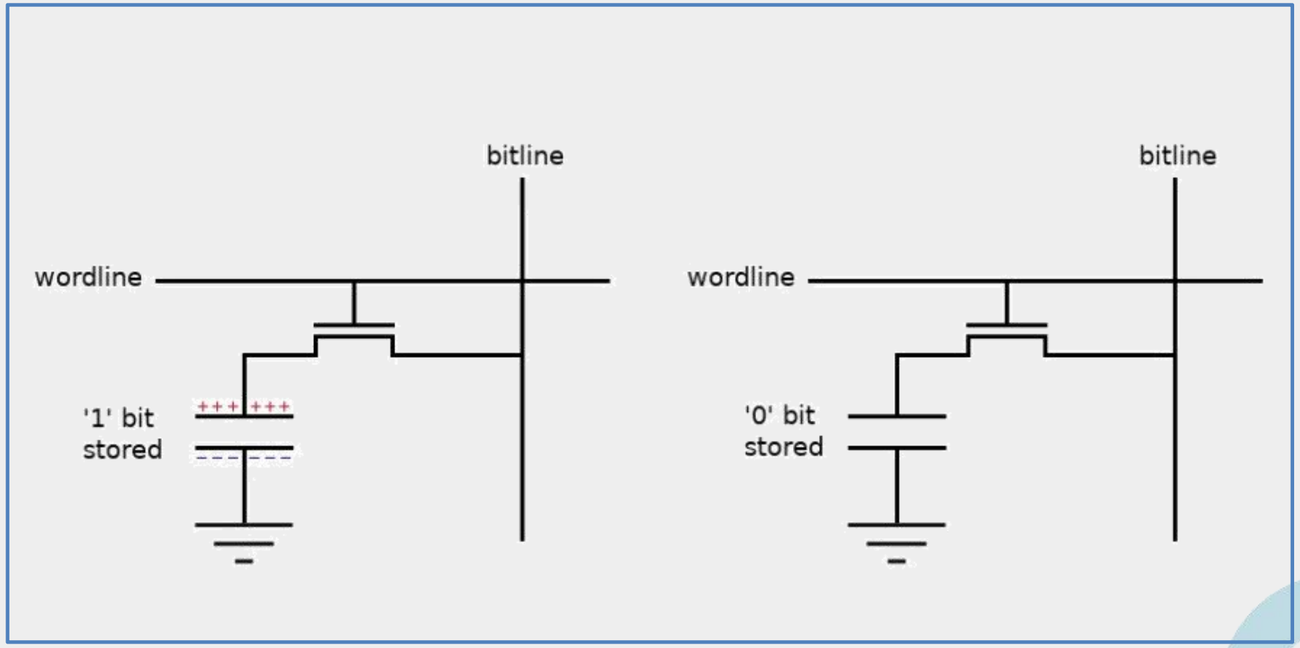
Dynamic Ram (DRAM)

How SRAM Works?



How DRAM Works?



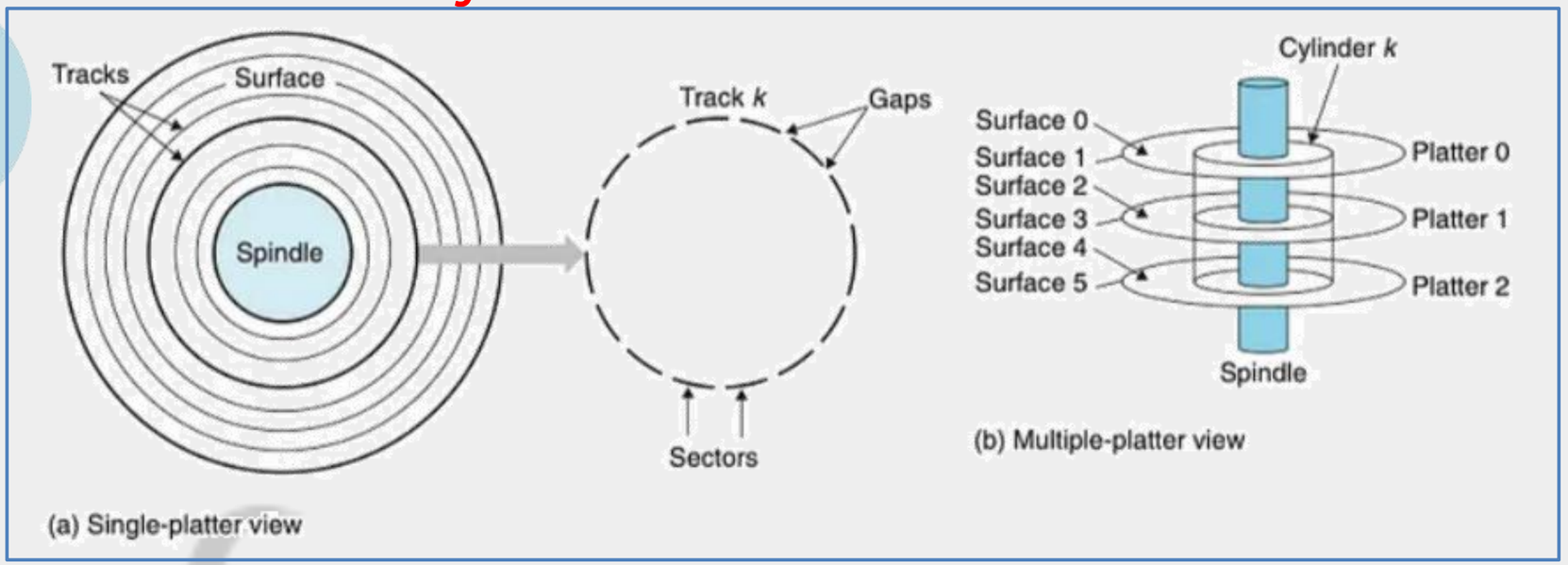


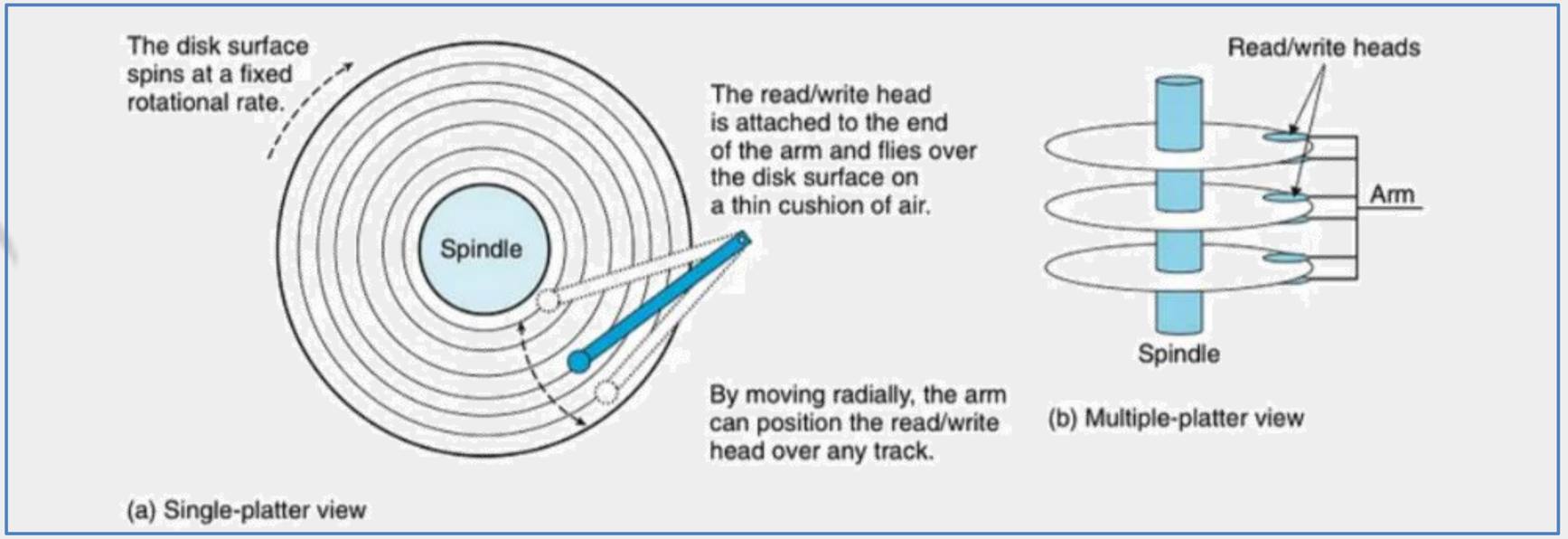
4. Secondary Storage

Feature	Details
Location	Internal or external storage device
Purpose	Permanent storage of data and programs
Speed	Slower than RAM
Capacity	Large
Volatility	Non-volatile
Cost per Bit	Low
Common Devices	NVMe SSD, SATA SSD, Hard Disk Drive (HDD)
Advantages	Large storage capacity, permanent storage, lower cost
Example	Windows OS, games, documents, and videos are stored on SSDs or HDDs.



Disk Geometry





5. Tertiary Storage

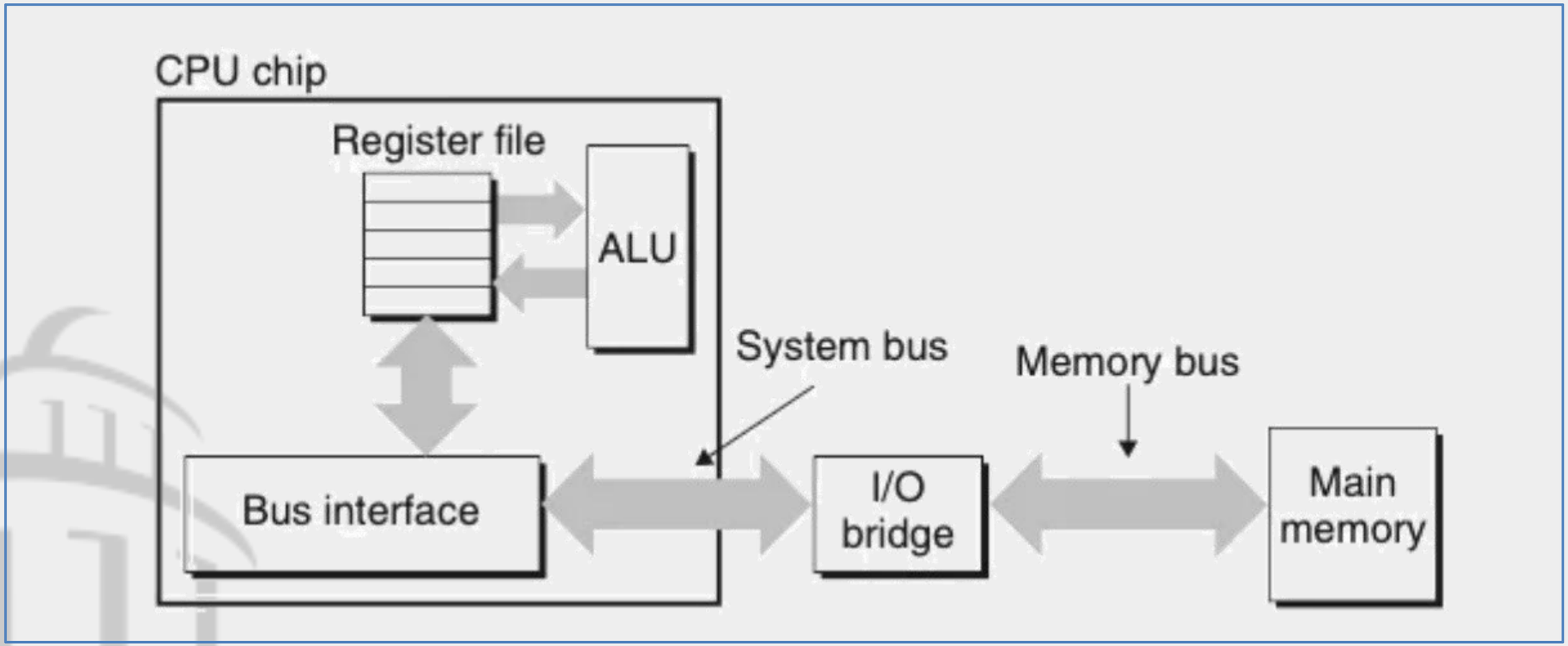
Feature	Details
Purpose	Long-term backup and archival storage
Speed	Slowest
Capacity	Very Large
Volatility	Non-volatile
Cost per Bit	Lowest
Common Devices	Magnetic Tape, Blu-ray Disc, Enterprise Backup Systems, Cold Cloud Storage
Applications	Data backup, government archives, enterprise storage, scientific data preservation
Example	Organizations use magnetic tape libraries for secure long-term backups.



Data Communication and Buses

- Communication between storage layers relies heavily on Buses.
- It consists of a collection of parallel wires that carry three distinct types of information:
    a) Address Bus: Carries the specific memory location ID.
    b) Data Bus: Carries the actual information being transferred.
    c) Control Bus: Carries commands (e.g., “Read,” “Write,” “Status,” “Clock”).

Data Communication and Buses

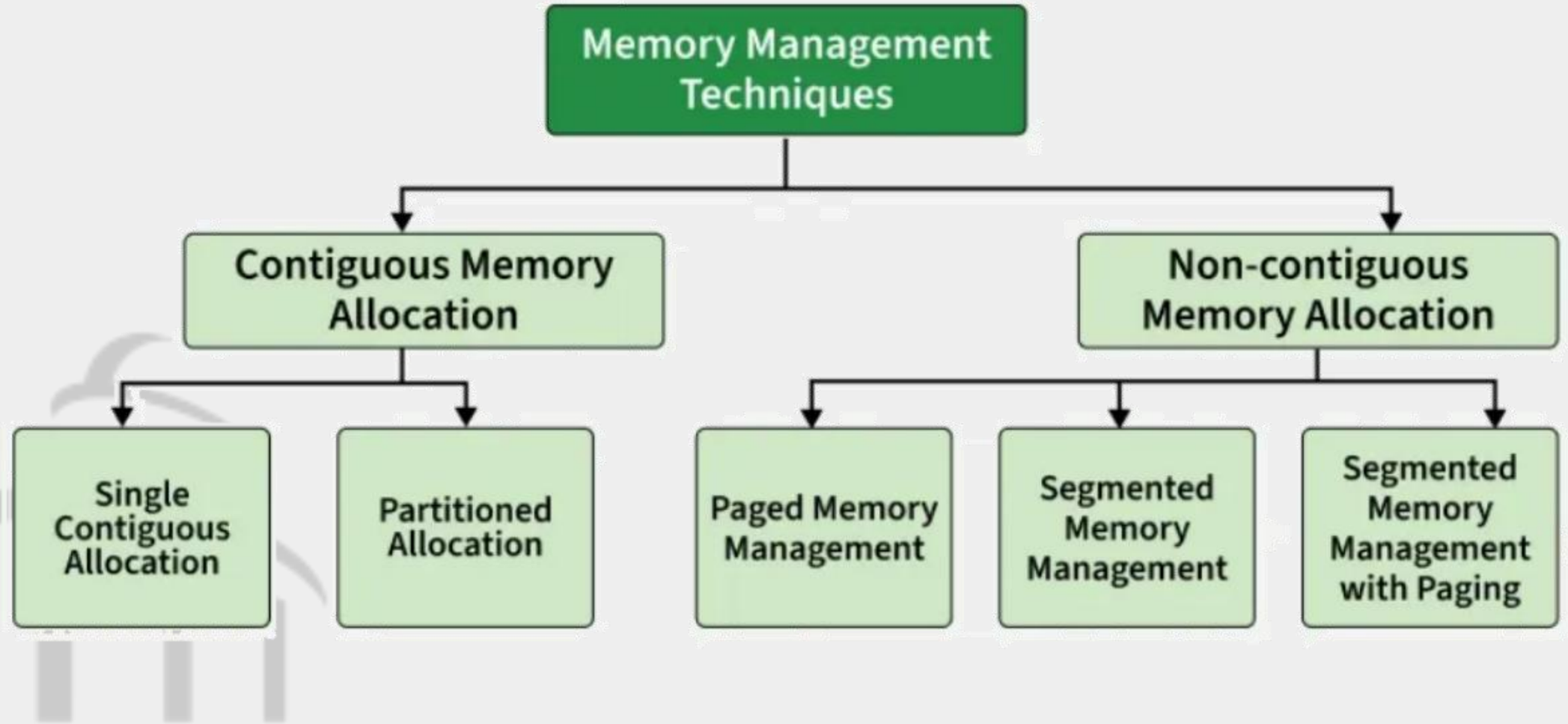


Applications of Memory Hierarchy

- Personal Computers
- Smartphones
- Cloud Computing
- Databases:
- Artificial Intelligence: Memory hierarchy is used by artificial intelligence for processing large data sets, which might be beyond the limit of fast memory storage.

04 Memory Allocation

- Memory allocation refers to the process of assigning memory space to programs and processes.
- OS uses this meachanism to allow multiprograms run simultaneously.
- It ensures efficient use of available memory while preventing conflicts between different processes.

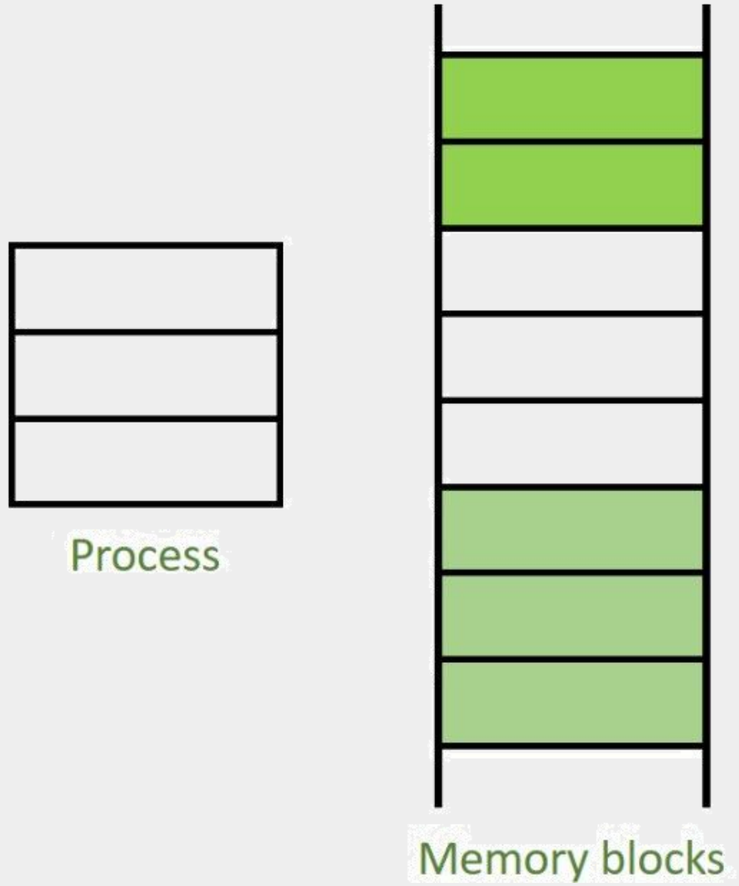

Memory Management Techniques
Contiguous Memory Allocation
Non-contiguous Memory Allocation
Single Contiguous Allocation
Partitioned Allocation
Paged Memory Management
Segmented Memory Management
Segmented Memory Management with Paging

a) Contiguous Memory Allocation

- Contiguous memory allocation is a memory management method where each process is given a single, continuous block of memory.
- This means all the data for a process is stored in adjacent memory locations.

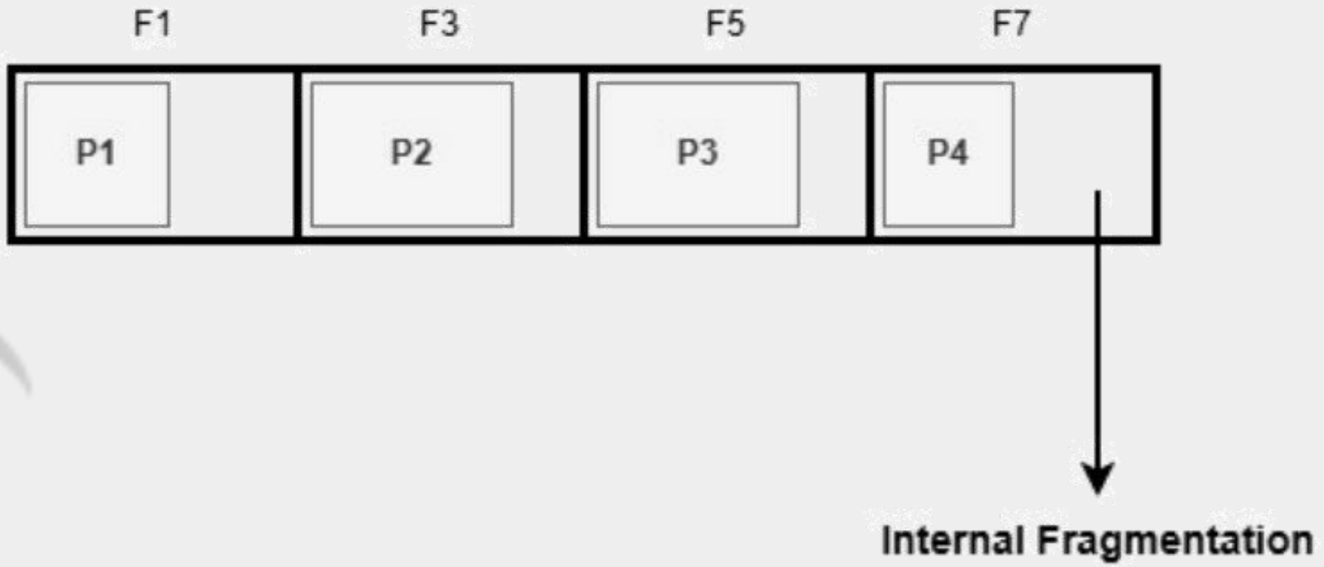
  i) Fixed Partitioning (Static)

  ii) Dynamic Partitioning (Variable)



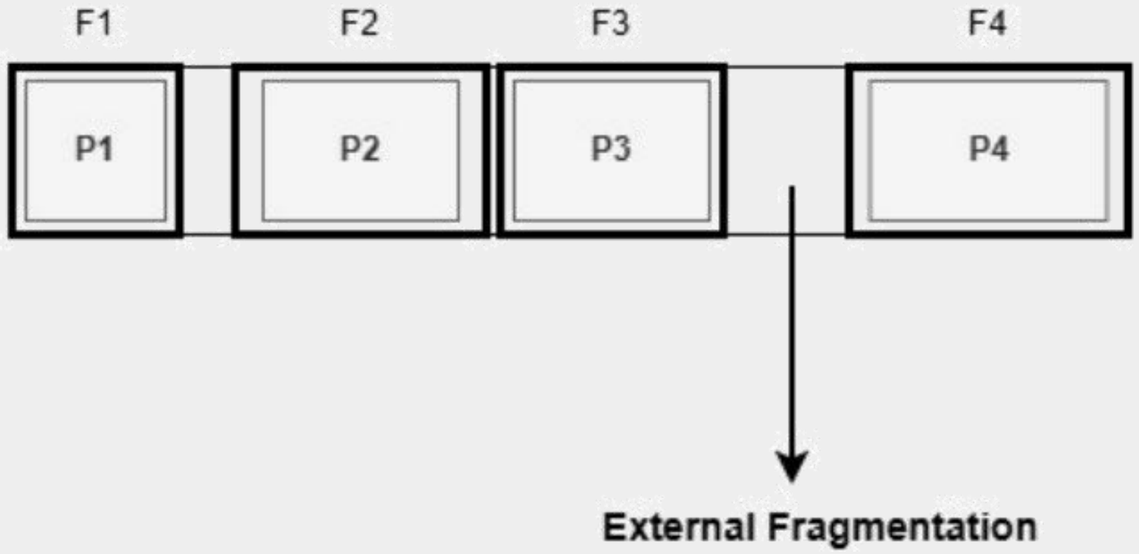
i) Fixed Partitioning (Static)

- Main memory is divided into partitions of fixed sizes.
- Each partition can hold exactly one process. When a new process loaded into memory, it is allocated a partition of fixed size.



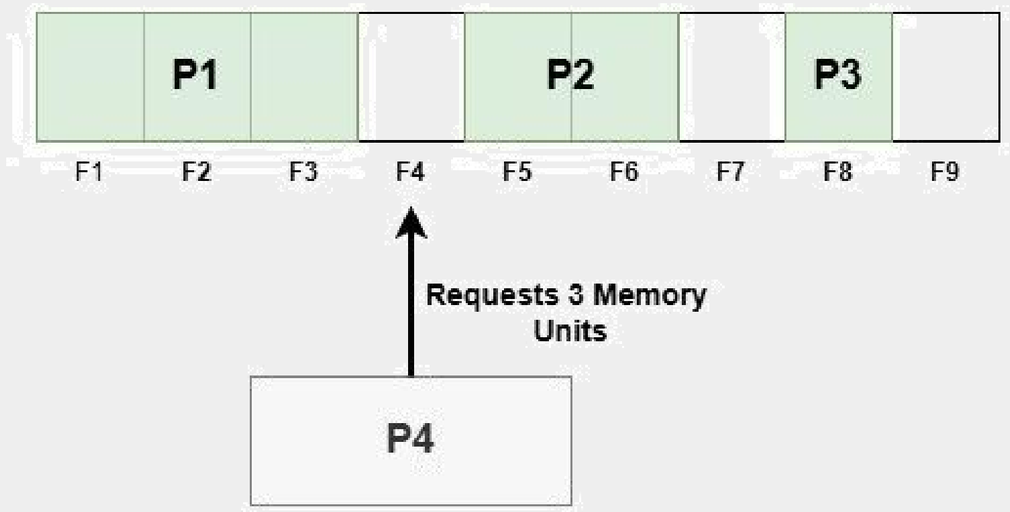
ii) Dynamic Partitioning or Variable Partioning

- Main memory is divided into partitions of varying sizes depending on the size of the processes being loaded into memory.
- When a new process is loaded, it is allocated to the first available partition that is large enough to hold the process.



Why Fragementation?

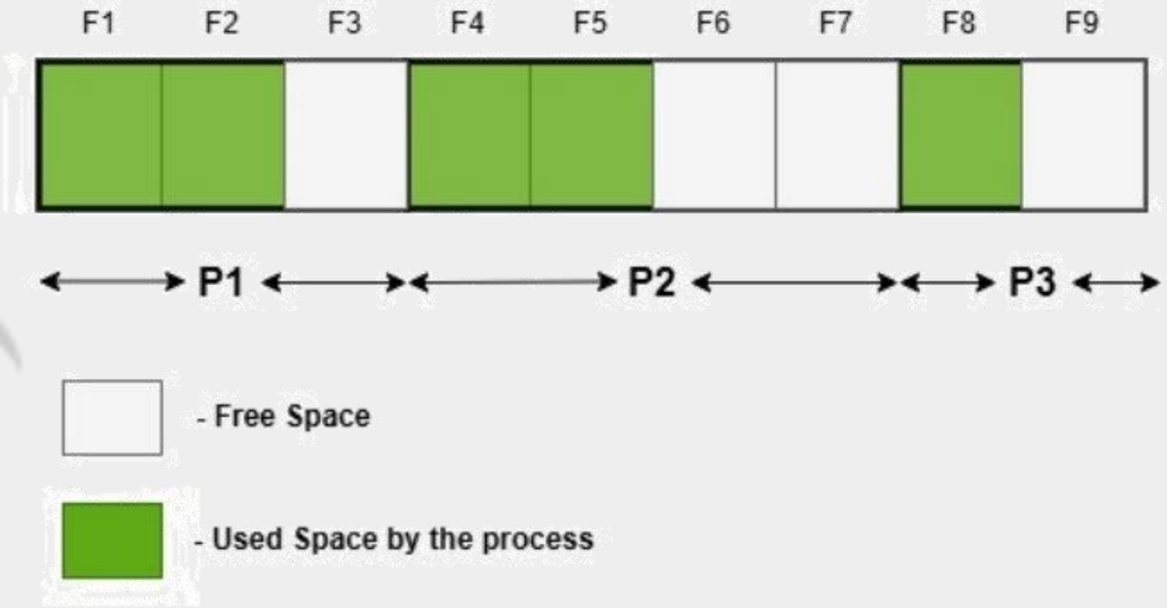
- Fragmentation is a condition where the available memory is divided into small and non-contiguous blocks after allocating and deallocating memory to various processes.
- Fragmentation = gap.
- Fragmentation can lead to wastage of memory.



Types of Fragementation

i) Internal Fragmentation

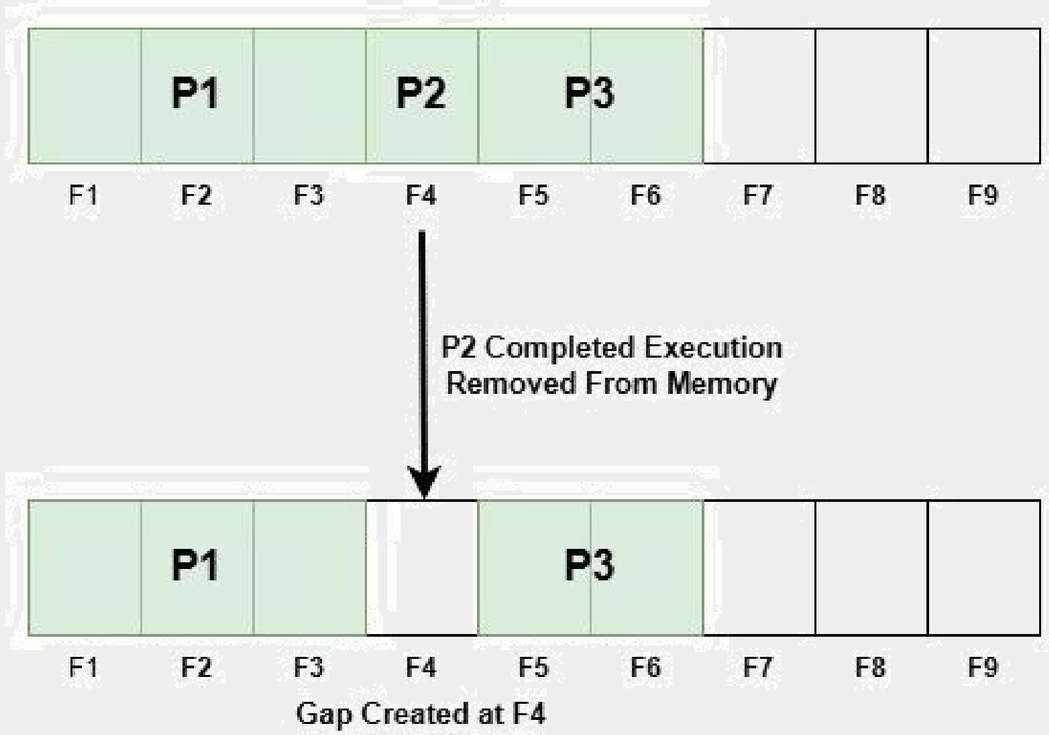
- Occurs when a process gets more memory than it actually needed.
- This can happen for large number of processes running in the system, which will lead to wastage of large amount of memory.



- Causes of Internal Fragmentation:
    a) Fixed Size Memory Allocation
    b) Management Overheads
    c) Memory Alignment

ii) External Fragmentation

- Refers to gaps generated in memory when multiple processes are loaded and unloaded from memory.

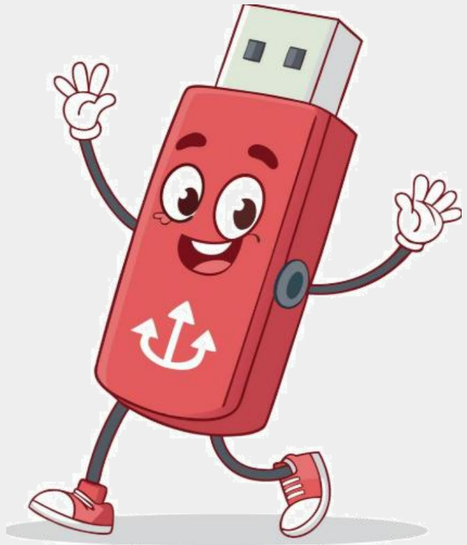


- Causes of Internal Fragmentation:
    a) Dynamic Memory Allocation
    b) Process Termination
    c) Variable Siza Memory Allocation

- How to reduce fragmentation?
    a) Compaction
    b) Paging
    c) Segmentation

Non-Contiguous Allocation

We’ll continue in Part B!



THANK YOU



